

SOFTWARE ENGINEERING LAB – LAB 3

Component Modelling & Architectural Pattern Selection

Hospital Bed & ICU Allocation Dashboard | Section H

Architecture Selection

We chose **Microservices Architecture** for the Hospital Bed & ICU Allocation Dashboard. The system is divided into independently deployable functional services for triage and ICU allocation, bed management, and ambulance transfer management, with a hospital data and authentication service.

Architectural Style Analysis

Style	Advantages	Limitations
Layered	Clear separation of presentation, business and data concerns; simple to understand.	Tight coupling between layers; difficult to scale individual components.
Microservices	Independent scaling and fault isolation; services can be deployed independently.	Operational complexity, network latency and data-consistency challenges.
Client-Server	Centralized control and simple deployment; straightforward data consistency.	Central server can become a bottleneck and single point of failure.

Why Microservices?

- 1. Independent scaling.** Bed availability and triage operations can receive high request volumes during busy periods. Individual services can be scaled without scaling the entire application.
- 2. Fault isolation.** Critical functions such as ICU prioritization and bed allocation can be isolated from failures in ambulance-transfer functionality, improving system resilience.

Security Advantage

Authentication is centralized through the Hospital Data & Authentication Service. Only authenticated users should access patient and bed-allocation information, while service-level access controls can restrict sensitive operations.

Performance Benefit

Frequently accessed services such as Bed Management and Triage & ICU Allocation can be independently scaled. This supports responsive bed-status retrieval and rapid triage/allocation decisions without requiring the complete system to scale together.

Component Model

Component	Responsibility
ER Dashboard	Hospital staff interface for viewing and initiating authorized operations.
Triage & ICU Allocation Service	Calculates triage severity and prioritizes ICU transfer requests.
Bed Management Service	Maintains current bed status and ward occupancy information.
Transfer Management Service	Initiates, tracks and escalates ambulance transfers.
Hospital Data & Authentication Service	Provides authenticated access and persistent patient, bed and transfer data.

Interfaces Shown in the Component Diagram

Interface	Provider	Consumer
Triage / ICU API	Triage & ICU Allocation Service	ER Dashboard
Bed Status API	Bed Management Service	ER Dashboard
Transfer API	Transfer Management Service	ER Dashboard
Patient & Vital Data API	Hospital Data & Authentication Service	Triage & ICU Allocation Service
Bed Data API	Hospital Data & Authentication Service	Bed Management Service
Transfer Data API	Hospital Data & Authentication Service	Transfer Management Service

UML notation: components use component notation; provided interfaces use ball notation and required interfaces use socket notation. Assembly connectors join the required interfaces to the corresponding provided interfaces.